

Multivariate Chain-Rule

In the multivariate case, where $\mathbf{x} \in \mathbb{R}^n$, the basic differentiation rules that we know from school (e.g., sum rule, product rule, chain rule) still apply. However, we need to pay attention because now we have to deal with matrices where multiplication is no longer commutative, i.e., the order matters.

$$\text{Product Rule: } \frac{\partial}{\partial \mathbf{x}} (f(\mathbf{x})g(\mathbf{x})) = \frac{\partial f}{\partial \mathbf{x}} g(\mathbf{x}) + f(\mathbf{x}) \frac{\partial g}{\partial \mathbf{x}} \quad (1)$$

$$\text{Sum Rule: } \frac{\partial}{\partial \mathbf{x}} (f(\mathbf{x}) + g(\mathbf{x})) = \frac{\partial f}{\partial \mathbf{x}} + \frac{\partial g}{\partial \mathbf{x}} \quad (2)$$

$$\text{Chain Rule: } \frac{\partial}{\partial \mathbf{x}} (g \circ f)(\mathbf{x}) = \frac{\partial}{\partial \mathbf{x}} (g(f(\mathbf{x}))) = \frac{\partial g}{\partial \mathbf{f}} \frac{\partial f}{\partial \mathbf{x}} \quad (3)$$

Product rule: $(fg)' = f'g + fg'$, Sum rule: $(f+g)' = f' + g'$, Chain rule: $(g \circ f)' = g'(f)f'$

Let us have a closer look at the chain rule. The chain rule formula (3) resembles to some degree the rules for matrix multiplication where “neighboring” dimensions have to match for matrix multiplication to be defined. If we go from left to right, the chain rule exhibits similar properties: ∂f shows up in the “denominator” of the first factor and in the “numerator” of the second factor. If we multiply the factors together, multiplication is defined (the dimensions of ∂f match, and ∂f “cancels”, such that $\partial g/\partial \mathbf{x}$ remains.¹

Consider a function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ of two variables x_1, x_2 . Furthermore, $x_1(t)$ and $x_2(t)$ are themselves functions of t . To compute the gradient of f with respect to t , we need to apply the chain rule (3) for multivariate functions as

$$\frac{df}{dt} = \begin{bmatrix} \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} \end{bmatrix} \begin{bmatrix} \frac{\partial x_1(t)}{\partial t} \\ \frac{\partial x_2(t)}{\partial t} \end{bmatrix} = \frac{\partial f}{\partial x_1} \frac{\partial x_1}{\partial t} + \frac{\partial f}{\partial x_2} \frac{\partial x_2}{\partial t} \quad (4)$$

where d denotes the gradient and ∂ partial derivatives.

Example:

Consider $f(x_1, x_2) = x_1^2 + 2x_2$, where $x_1 = \sin t$ and $x_2 = \cos t$, then

$$\frac{df}{dt} = \frac{\partial f}{\partial x_1} \frac{\partial x_1}{\partial t} + \frac{\partial f}{\partial x_2} \frac{\partial x_2}{\partial t} \quad (5)$$

$$= 2 \sin t \frac{\partial \sin t}{\partial t} + 2 \frac{\partial \cos t}{\partial t} \quad (6)$$

$$= 2 \sin t \cos t - 2 \sin t = 2 \sin t (\cos t - 1) \quad (7)$$

is the corresponding derivative of f with respect to t .

If $f(x_1, x_2)$ is a function of x_1 and x_2 , where $x_1(s, t)$ and $x_2(s, t)$ are themselves functions of two variables s and t , the chain rule yields

$$\frac{df}{ds} = \frac{\partial f}{\partial x_1} \frac{\partial x_1}{\partial s} + \frac{\partial f}{\partial x_2} \frac{\partial x_2}{\partial s}, \quad (8)$$

¹This is only an intuition, but not mathematically correct since the partial derivative is not a fraction.

$$\frac{df}{dt} = \frac{\partial f}{\partial x_1} \frac{\partial x_1}{\partial t} + \frac{\partial f}{\partial x_2} \frac{\partial x_2}{\partial t}, \quad (9)$$

which can be expressed as the matrix multiplication

$$\frac{df}{d(s, t)} = \frac{\partial f}{\partial \mathbf{x}} \frac{\partial \mathbf{x}}{\partial (s, t)} = \underbrace{\begin{bmatrix} \frac{\partial f}{\partial x_1} & \frac{\partial f}{\partial x_2} \end{bmatrix}}_{=\frac{\partial f}{\partial \mathbf{x}}} \underbrace{\begin{bmatrix} \frac{\partial x_1}{\partial s} & \frac{\partial x_1}{\partial t} \\ \frac{\partial x_2}{\partial s} & \frac{\partial x_2}{\partial t} \end{bmatrix}}_{=\frac{\partial \mathbf{x}}{\partial (s, t)}}. \quad (10)$$

This compact way of writing the chain rule as a matrix multiplication makes only sense if the gradient is defined as a row vector. Otherwise, we will need to start transposing gradients for the matrix dimensions to match. This may still be straightforward as long as the gradient is a vector or a matrix; however, when the gradient becomes a tensor (we will discuss this in the following), the transpose is no longer a triviality.

Example: (Gradient of a Linear Model)

Let us consider the linear model

$$\mathbf{y} = \Phi \boldsymbol{\theta}, \quad (11)$$

where $\boldsymbol{\theta} \in \mathbb{R}^D$ is a parameter vector, $\Phi \in \mathbb{R}^{N \times D}$ are input features and $\mathbf{y} \in \mathbb{R}^N$ are the corresponding observations. We define the following functions:

$$L(e) := \|e\|^2, \quad (12)$$

$$e(\boldsymbol{\theta}) := \mathbf{y} - \Phi \boldsymbol{\theta}. \quad (13)$$

We seek $\frac{\partial L}{\partial \boldsymbol{\theta}}$, and we will use the chain rule for this purpose.

Before we start any calculation, we determine the dimensionality of the gradient as

$$\frac{\partial L}{\partial \boldsymbol{\theta}} \in \mathbb{R}^{1 \times D}. \quad (14)$$

The chain rule allows us to compute the gradient as

$$\frac{\partial L}{\partial \boldsymbol{\theta}} = \frac{\partial L}{\partial \mathbf{e}} \frac{\partial \mathbf{e}}{\partial \boldsymbol{\theta}}. \quad (15)$$

We know that $\|e\|^2 = \mathbf{e}^\top \mathbf{e}$ and determine

$$\frac{\partial L}{\partial \mathbf{e}} = 2\mathbf{e}^\top \in \mathbb{R}^{1 \times N}. \quad (16)$$

Furthermore, we obtain

$$\frac{\partial \mathbf{e}}{\partial \boldsymbol{\theta}} = -\Phi \in \mathbb{R}^{N \times D}, \quad (17)$$

such that our desired derivative is

$$\frac{\partial L}{\partial \boldsymbol{\theta}} = -2\mathbf{e}^\top \Phi \stackrel{(13)}{=} -\underbrace{2(\mathbf{y}^\top - \boldsymbol{\theta}^\top \Phi^\top)}_{1 \times N} \underbrace{\Phi}_{N \times D} \in \mathbb{R}^{1 \times D}. \quad (18)$$

Remark. We would have obtained the same result without using the chain rule by immediately looking at the function

$$L_2(\boldsymbol{\theta}) := \|\mathbf{y} - \Phi\boldsymbol{\theta}\|^2 = (\mathbf{y} - \Phi\boldsymbol{\theta})^\top (\mathbf{y} - \Phi\boldsymbol{\theta}). \quad (19)$$

This approach is still practical for simple functions like L_2 but becomes impractical if consider deep function compositions.
